

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech. Branch :Electronics & Telecommunication Engineering

Semester :VII Subject Code & Name: BTETPE703A Embedded System Design

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.		12
A) Provide a brief explanation of six fundamental blocks of an embedded system and analyze their roles and interactions.	CO1	6
B) Identify and describe the key characteristics of an embedded system.	CO1	6
C) With a neat block diagram explain the differences between Von Neumann and Harvard Architectures.	Level 3	6
Q.2 Solve Any Two of the following.		12
A) Briefly explain following types of memory. DRAM EPROM Flash Memory	CO2	6
B) Briefly explain the classification of processors used in Embedded System.	CO2	6
C) Explain in brief about following non-functional requirements needs to be specified before designing an Embedded System. Performance, Cost, & Power Consumption.	CO3	6
Q.3 Solve Any Two of the following.		12
A) Briefly explain different steps of signal conditioning.	Level 4	6
B) Explain in brief about I2C bus interfacing with microprocessor / microcontroller.	CO2	6
C) Explain in brief about Analog to Digital Converter interfacing with microprocessor / microcontroller.	Level 1	6

Q.4 Solve Any Two of the following.		12
A) Distinguish between Hard Real Time Systems and Soft Real Time System.	Level 2	6
B) Explain in brief Task/Process Scheduling, Task/Process Synchronization & Error/Exception handling.	Level 2	6
C) Briefly Explain any six characteristics of Real-Time System.	CO4	6
Q.5 Solve Any Two of the following.		12
A) What are Rate Monotonic and Deadline Monotonic scheduling? State any four differences between them.	CO4	6
B) What are the advantages and limitations of Earliest Deadline First scheduling algorithm?	Level 2	6
C) What are the characteristics and limitations of Kernel-Level Threads.	Level 2	6

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